

Low-Cost Edge-AI IoT Node for Algal Bloom Prediction

Research proposal and honest feasibility assessment · builds on the reproduction and leakage audit of Zhou et al. (2026), PLOS ONE 21(6): e0351325

The idea in one sentence: a solar-powered sensor node costing about **\$50** measures water temperature, light and chlorophyll in a pond, lake or fish farm, and a compressed machine-learning model running **on its \$5 chip** forecasts chlorophyll-a (algae) so owners get an early warning of algal blooms — with every model evaluated in a strictly leakage-free way.

1. Why it matters

- **Algal blooms** kill fish, damage fish farms, and make drinking and recreational water unsafe. Chlorophyll-a is the standard indicator of algae.
- **Monitoring is expensive:** commercial chlorophyll sensors and monitoring buoys typically cost thousands of US dollars, so most small water bodies and farms in low-income regions are not monitored at all.
- **Published ML results are often too optimistic:** data leakage has been documented in 294 papers across 17 fields (Kapoor & Narayanan, 2023). We found it in a 2026 marine benchmark too. A cheap device is only useful if its predictions are honestly tested.

2. What we already have (starting evidence)

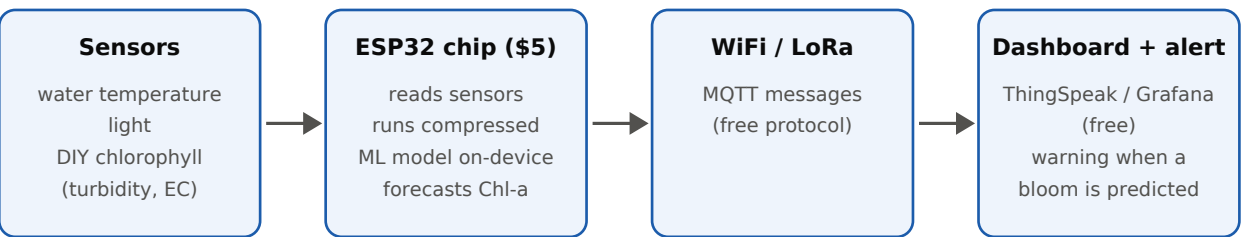
Result	Evidence
Exact reproduction of a 2026 PLOS ONE benchmark	52/52 reported values and 7/7 p-values matched
Two data-leakage errors found in that paper	LSTM models trained on 84–86% of their test data; rolling features containing the target
Better chlorophyll model on CTD profile data	R^2 0.458 → 0.633 (+38%), significant, same test rows
Open code and results	github.com/ali-Hamza817/ML-Models-for-Marine-and-Atmospheric-Environmental-Prediction

Cheap-sensor test (hydrographic chlorophyll data, one tuned LightGBM, same split):

Sensors available to the model	Test R^2 (train+val fit)	What it means
All sensors incl. dissolved oxygen + chlorophyll history	0.576	Reference: lab-grade sensor set
Low-cost sensors + chlorophyll history (no DO sensor)	0.552	Oxygen probe (~\$150+) is not needed
Low-cost sensors only (no DO, no chlorophyll sensor)	-0.210	Chlorophyll sensor is essential
Temperature + light + chlorophyll history only	0.534	Minimal cheap node still works

Caution: in this data the chlorophyll history comes from a lab-grade sensor. A DIY sensor is noisier, so the real accuracy on the node must be measured, not assumed.

3. The proposed system



Part	Purpose	Cost (USD)
ESP32 development board	Sensor reading, on-device ML, WiFi	5
DS18B20 waterproof temperature probe	Water temperature	2
BH1750 light sensor in clear sealed housing	Sunlight (replaces PAR sensor)	2
DIY fluorometer: blue LED (~470 nm) + photodiode + red filter	Chlorophyll-a (red fluorescence)	5–10
Turbidity sensor, TDS/EC probe (optional)	Water clarity, salinity	14
18650 battery, TP4056 charger, small solar panel	Power	8
Waterproof box, cable glands	Housing	5
Total	Software is free (Arduino, MQTT, ThingSpeak/Grafana, emlearn, TFLite Micro)	≈ 40–55

4. Research questions

#	Research question	How it is answered	Evidence so far
RQ1	How much forecasting accuracy is lost when lab-grade inputs are replaced by low-cost sensors?	Sensor-ablation on public data, then the same comparison on the node's own field data	Dropping the oxygen probe: R^2 0.576 → 0.552
RQ2	Can a leakage-free chlorophyll forecaster be compressed to run on a \$5 microcontroller without significant accuracy loss?	Tree pruning, fewer/shallower trees, int8-quantised GRU, distillation; measure R^2 , memory, latency and energy on a real ESP32	Current models are too big: LightGBM ≈ 2.1 MB, GRU ensemble ≈ 3.7 MB vs ESP32 520 KB RAM / 4 MB flash
RQ3	How accurate is a ~\$10 DIY fluorometer against laboratory chlorophyll-a measurements, and can ML correct its temperature and fouling drift?	Dilution series of algae culture + lab (acetone extraction) samples; calibration models; drift tracked over weeks	Open question — highest-risk part
RQ4	Do models trained on public benchmark data transfer to a new local water body, and how much local data is needed?	Zero-shot vs fine-tuned with 1, 2, 4, 8 weeks of local data; strictly chronological tests	Open question

5. Is it novel? What already exists

A quick literature search found close prior work. **Each building block already exists**; the novelty must come from combining them with rigorous evaluation and real field validation.

Existing work	What it did	Gap this project fills
Leeuw et al., 2013 (Sensors) — low-cost Arduino fluorometer	In-situ phytoplankton fluorescence for about \$150	No forecasting, no on-device ML; we target ~\$10 optics
Mozo et al., 2022 (Scientific Reports) — chlorophyll soft-sensor	Random Forest predicts Chl-a from temperature, pH, EC; designed for buoys; chronological evaluation	Models proposed for buoys but not run on a microcontroller; used existing monitoring stations
KICT, 2025 (Environmental Monitoring and Assessment)	Cheap light sensors + SVM/regression label algal blooms and estimate Chl-a	Detects the current state; no forecasting, no leakage-aware evaluation
Low-cost IoT HAB device with spectrofluorometry + imaging (Springer, 2025)	Integrated early-warning hardware with ML anomaly detection	More complex/costly optics; not a \$50 TinyML node
TinyML for environmental prediction (e.g. ozone 2025, irrigation 2026)	Forecasting models on edge devices	Not applied to chlorophyll/algal blooms
Kapoor & Narayanan, 2023 (Patterns) — leakage in ML science	Leakage found in 294 papers, 17 fields	We add a documented marine case and a leakage-free protocol for IoT forecasting

Defensible novelty claim: the first end-to-end, leakage-audited pipeline that takes a published marine benchmark, fixes its evaluation, compresses the improved forecaster onto a \$5 microcontroller, and validates it on a ~\$50 node against laboratory chlorophyll measurements. **Before writing “first”, run a proper systematic search in Scopus / Web of Science** — this check was quick and could have missed papers.

6. Is it enough for a Q1 journal? (honest answer)

What you deliver	Realistic outcome
Only a sensor node + standard ML on a few weeks of own data, no lab reference	Q1 unlikely — very common topic; conference or Q3/Q2
Only model compression on public datasets (no hardware field test)	Q2 possible, Q1 unlikely
Leakage audit + improved benchmark (work already done), written carefully	Q1/Q2 possible as a reproducibility / methods paper
Full package: leakage-free models + measured on a real ESP32 (latency, energy) + DIY sensor calibrated against lab samples + ≥ 3 months field data + open hardware and code	Realistic Q1 chance — still decided by reviewers, never guaranteed

Candidate journal	Fit	Quartile
Computers and Electronics in Agriculture (Elsevier)	Best fit if applied to fish ponds / aquaculture	Q1 (Scimago 2024, SJR 2.165)
Environmental Modelling & Software (Elsevier)	Leakage audit, benchmark and open software	Q1 (Scimago 2024, SJR 1.466)
Ecological Informatics (Elsevier)	Ecological sensing + ML	Check on scimagojr.com
IEEE Internet of Things Journal	IoT system + TinyML (very competitive)	Check on scimagojr.com
HardwareX (Elsevier)	Open-source DIY fluorometer / node design	Check on scimagojr.com
Sensors (MDPI)	Realistic backup for the node paper	Check on scimagojr.com

Suggested strategy: **two papers**. Paper A = leakage audit + improved benchmark (almost ready). Paper B = the IoT node + TinyML + field validation. Two linked papers make a stronger PhD application than one.

7. Is it worth doing?

For	Against / risk	How to reduce the risk
Very low cost (≈ \$50)	DIY fluorometer may be too noisy	Calibrate with algae dilution series and a few lab samples (often free at a university lab)
Strong starting point already done	Field data takes months	Start bottle experiments (algae + fertiliser in sunlight) in week 1 while waiting
Combines hardware, ML and rigorous evaluation — skills PhD panels value	Sensor fouling in water	Weekly cleaning, copper tape, log cleaning dates
Real-world impact (fish farms, water safety)	Few natural bloom events	Nutrient-enriched containers create controlled blooms
Open-source portfolio on GitHub	Journal rejection	Keep a Q2 backup journal; publish data/code openly

Verdict: worth doing — if you can commit about **6-9 months** and can access (1) a water body or outdoor containers and (2) a lab for a few reference chlorophyll measurements. **Without lab reference measurements, do not target Q1**. A PhD scholarship is never guaranteed by one paper, but a real device, open code and honest, leakage-free results make a strong application.

8. Plan (about 7-9 months)

Months	Work	Output
1	Write Paper A (leakage audit + improvement); email the original authors	Paper A submitted
1-2	RQ2: compress models on the cluster; test on one ESP32 (latency, memory, energy)	Edge-AI results
2	Build node; RQ3 bottle experiments and lab calibration of the DIY fluorometer	Calibrated sensor
3-6	Field deployment (pond / fish farm / lake); weekly cleaning and lab spot-checks	≥ 3 months of data
6-7	RQ1 and RQ4 analysis with strictly chronological tests	Final results
7-9	Write Paper B; release hardware design, firmware, data and code	Paper B submitted

9. Minimum results needed before submitting Paper B

- DIY fluorometer vs lab chlorophyll-a: calibration R^2 and error reported honestly, including drift over time.
- At least 3 months of continuous field data, with gaps and cleaning events logged.
- The model actually running on the ESP32, with measured memory, inference time and battery life.
- Chronological train/validation/test split with no overlapping history windows (the error found in the 2026 paper).
- Comparison against simple baselines (persistence, mean) and the original paper's model types.
- All code, firmware, hardware design and data published openly.

10. Honesty rules

- Report only values measured by the node or computed from real data; never fill gaps with invented readings.
- Credit Zhou et al. (2026) and their open data; contact them about the leakage findings before publishing.
- Describe the deployment exactly as done (a pond is not an ocean), and report failures and negative results.

References

Zhou X, et al. (2026). Cross-dataset benchmarking of machine learning models for marine and atmospheric environmental prediction. PLOS ONE 21(6): e0351325. <https://doi.org/10.1371/journal.pone.0351325>

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TinyML ozone prediction (2025) <https://arxiv.org/abs/2504.03776> · TinyML precision irrigation (2026) <https://arxiv.org/abs/2601.13054> · Quartiles: SCImago 2024 (scimagojr.com) — re-check before submission.